3.1 Unit 1 PSYB1 Introducing Psychology

Students should have experience of designing and conducting informal classroom research using a variety of methods. They will be expected to analyse data collected in investigations and draw conclusions based on research findings. They will be required to draw on these experiences to answer questions in the examination for this unit.

Approaches

Topics	3.1.1 Key Approaches in Psychology 3.1.2 Biopsychology
Aims	• To introduce students to the key approaches in psychology. • To promote an understanding of the role of physiology in behaviour. • To promote an understanding of the genetic basis of behaviour. • To enable appreciation of research methods and ethical issues in the different approaches. • To develop an appreciation of how science works in psychology.
3.1.1 Key Approaches in Psychology	The basic assumptions of the following approaches: biological; behaviourist; social learning theory; cognitive; psychodynamic and humanistic. The distinguishing features of each approach, including research methods used. Biological: the influence of genes; biological structures; the evolution of behaviour. Behaviourist: classical conditioning; operant conditioning. Social Learning Theory: modelling; mediating cognitive factors. Cognitive: the study of internal mental processes and the use of models to explain these processes. Psychodynamic: the role of the unconscious; psychosexual stages; the structure of personality; defence mechanisms. Humanistic: free will; concepts of self and self-actualisation; conditions of worth. The strengths and limitations of each approach, including research methods used.
3.1.2 Biopsychology	
Physiological psychology	Basic understanding of the structure and function of neurons and synaptic transmission. The divisions of the nervous system. Localisation of function in the brain (cortical specialisation) including motor, somatosensory, visual, auditory and ‘language’ centres. Methods used to identify areas of cortical specialisation, including neurosurgery, post-mortem examinations; EEGs, electrical stimulation, scanning techniques, including PET. Actions of the sympathetic and parasympathetic divisions of the autonomic nervous system including, the adrenal glands, adrenalin and fight or flight response.
The genetic basis of behaviour	Difference between genotype and phenotype. Types of twins: monozygotic (MZ) and dizygotic (DZ). Use of twin studies, and family and adoption studies to investigate the genetic basis of behaviour.

Gender Development

Topic	3.1.3 Gender Development
Aims	• To demonstrate how key approaches can be applied to the development of gender. • To demonstrate how psychology provides an understanding of human development. • To develop an appreciation of how science works in relation to the investigation of gender development.
Outcomes	Upon completion of this topic, students should be able to • explain concepts related to the development of gender; • understand and appreciate the biological, social learning, cognitive and psychodynamic explanations of gender development.
3.1.3 Gender Development	
Concepts	Sex and gender: androgyny; sex-role stereotypes; cultural variations in gender-related behaviour; nature and nurture.
Explaining gender development	Biological explanations: typical and atypical sex chromosome patterns, including Klinefelter's syndrome and Turner's syndrome; influence of androgens (including testosterone) and oestrogens. Social learning theory: reinforcement; modelling; imitation and identification. Cognitive approach: Gender schema theory; Kohlberg's cognitive-developmental theory, including gender identity, gender stability and gender constancy. Psychodynamic approach: Freud's psychoanalytic theory; Oedipus complex; Electra complex; identification.

Research Methods

Topics	3.1.4 Methods of Research 3.1.5 Representing Data and Descriptive Statistics 3.1.6 Ethics
Aims	• To promote a critical understanding of quantitative and qualitative methods employed in psychological research. • To promote an understanding of the use of descriptive statistics. • To demonstrate how data can be represented. • To develop an awareness of ethical issues in psychological research. • To develop an appreciation of how science works in psychological research.
3.1.4 Methods of Research	
Planning Research	Qualitative and quantitative research: the distinction between qualitative and quantitative data collection techniques; strengths and limitations of quantitative and qualitative data. Formulating research questions. Stating aims. Formulating hypotheses (null and experimental/alternative/research). Populations and sampling. Sampling techniques, including opportunity, random, stratified and systematic.
Experimental Methods	Experiments: field, laboratory and quasi-experiments. Issue of ecological validity. Independent and dependent variables. Manipulation and control of variables in experiments. Extraneous and confounding variables.

	Experimental designs: repeated or related measures, matched pairs, independent groups and appropriate use of each.
	Controls associated with different designs, including counterbalancing. Strengths and limitations of different experimental designs.
	Strengths and limitations of experimental methods.
Non-experimental Methods	Self-report methods: questionnaire construction, including open and closed questions; types of interviews: structured and unstructured.
	Pilot studies and their value.
	Correlation studies. The difference between an experiment and a correlation study.
	Observational studies: natural and laboratory settings; covert and overt; participant and non-participant observation.
	The process of content analysis.
	Case studies. The role of case studies in psychology.
	Strengths and limitations of these methods.
3.1.5 Representing Data and Descriptive Statistics	
Representing Data	Appropriate use of the following tabular and graphical displays: bar charts, histograms, graphs, scattergrams and tables.
Descriptive Data	Calculation and use of measures of central tendency (mean, median, mode) and measures of dispersion (range and standard deviation).
	Correlation as a description of the relationship between two variables. Positive, negative and zero correlations.
3.1.6 Ethics	An awareness of the code of ethics in psychology as specified by the British Psychological Society.
	The application of the code of ethics in psychological research.